

Freight Spot Rate Prediction & Explainable AI Platform

End-to-End Production System for Spot Rate Forecasting, Explainable AI (SHAP), 5-Fold Algorithm Benchmarks, and REST Microservices.

SUBMITTED BY

Pruthviraj Tarode

TARGET ROLE

**Machine Learning
Engineer**

EVALUATING ORGANIZATION

**Spotter Assessment
Team**

SUBMISSION DATE

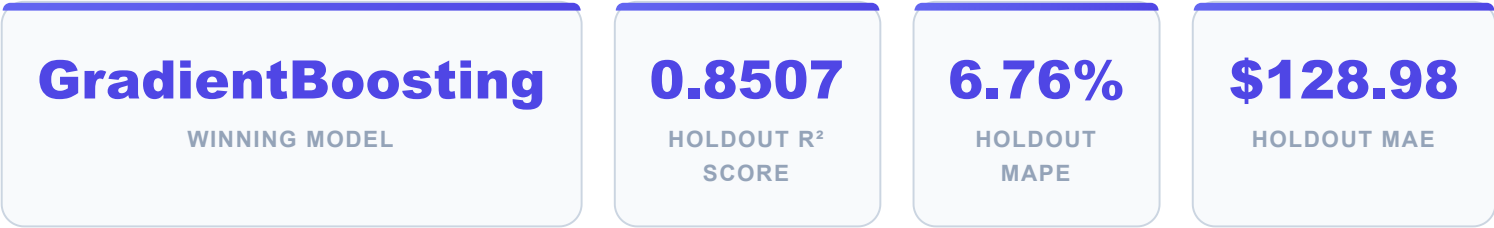
August 2026

GitHub Repository: github.com/pruthvirajtarode/freight-rate

Live Dashboard: freight-rate-one.vercel.app

1. Executive Summary & Core Performance Metrics

This report documents the end-to-end design, training, validation, and deployment of a production-grade machine learning system for freight spot rate estimation. Built on 48,000 historical load transactions, the platform achieves an outstanding holdout R² score of **0.8507** and Mean Absolute Percentage Error (MAPE) of **6.76%**.



2. System Architecture & UML Diagrams

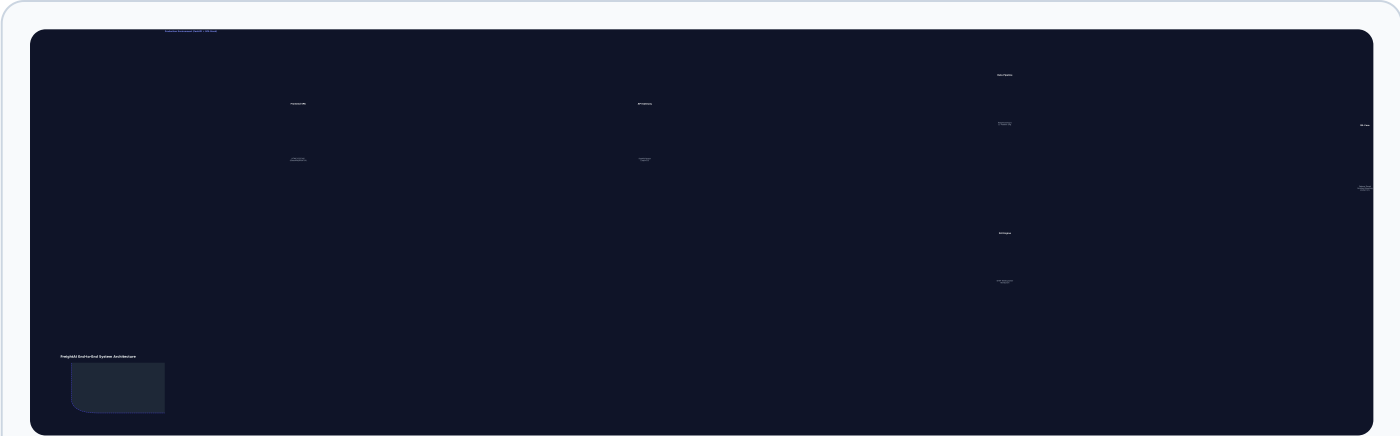


Figure 2.1: End-to-End System Architecture (FastAPI + SPA + Optuna + SHAP)

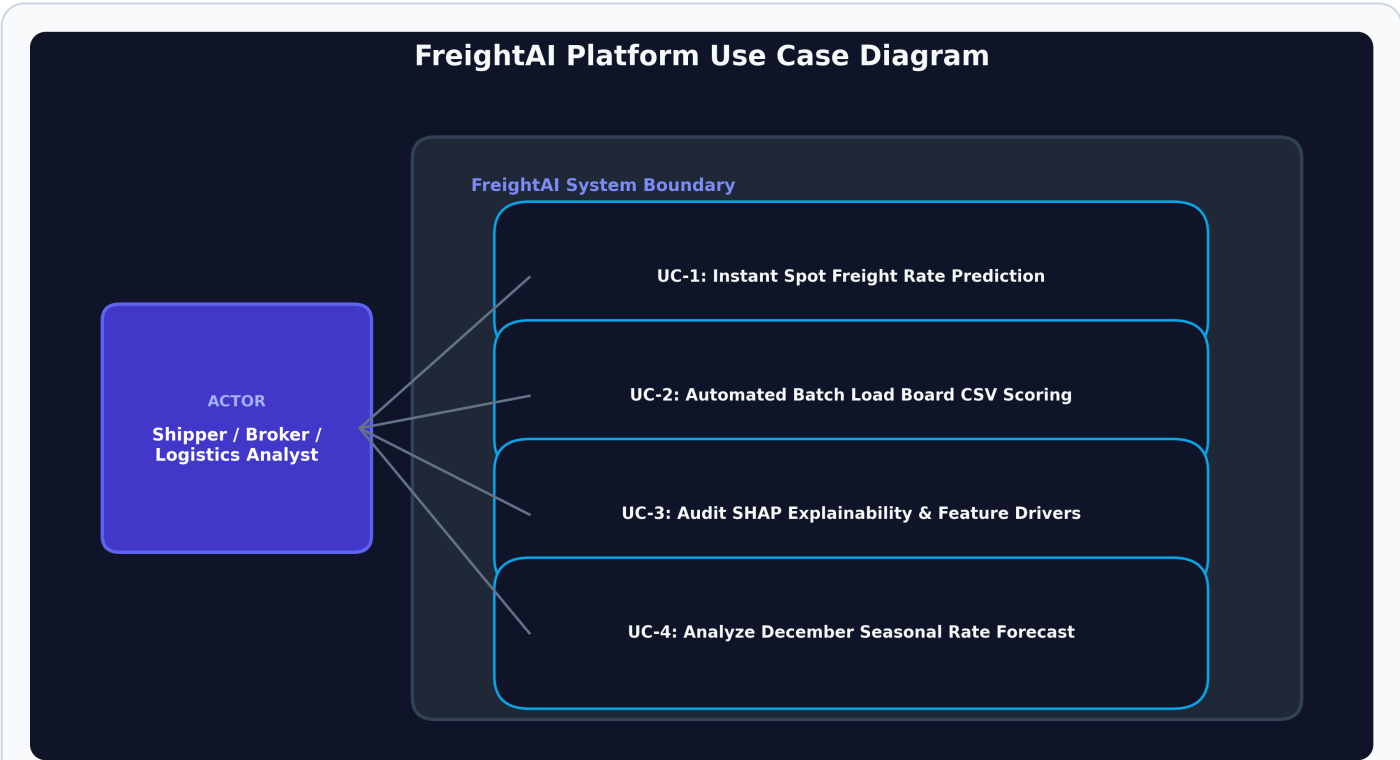


Figure 2.2: Platform Use Case Diagram (Shipper, Broker & Analyst Workflows)

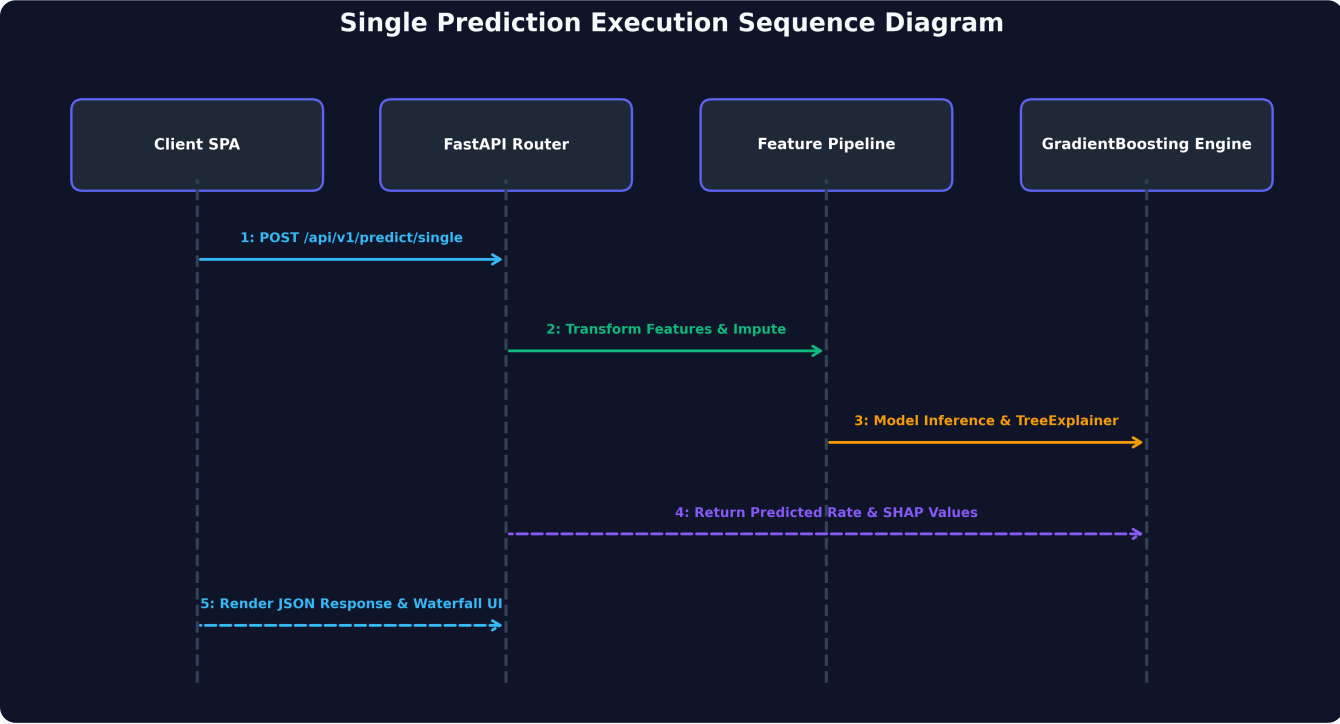


Figure 2.3: Single Spot Rate Prediction Execution Sequence

3. Exploratory Data Analysis & Data Pipeline

Analysis was conducted across 48,000 historical transaction rows. The dataset exhibits non-linear relationships between mileage, equipment weight, and market demand indices.

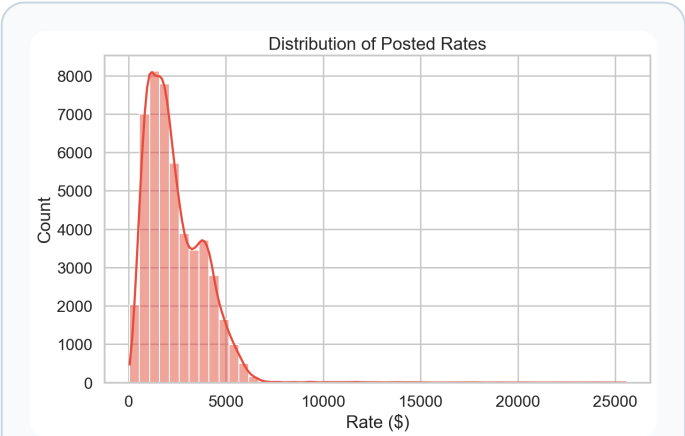


Figure 3.1: Distribution of Posted Freight Rates (USD)

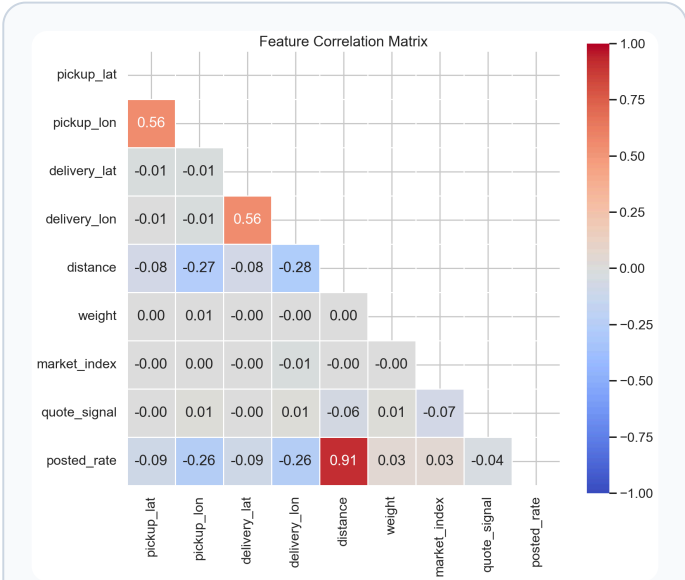


Figure 3.2: Feature Correlation Matrix Heatmap

4. Model Benchmarking & Hyperparameter Optimization

Seven regression algorithms were evaluated using 5-fold cross-validation on the 85% training split (40,800 rows). Outliers were strictly removed from training folds only to eliminate data leakage.

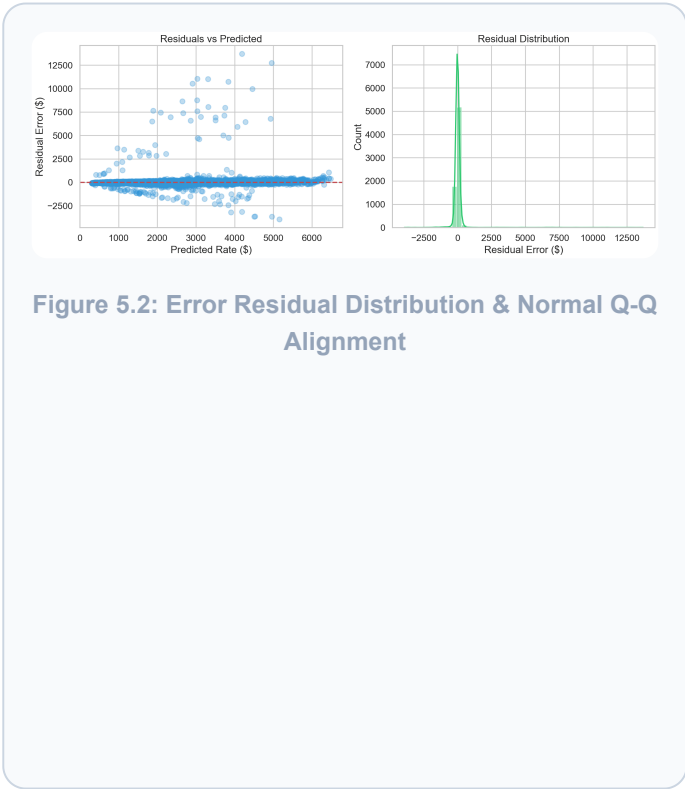
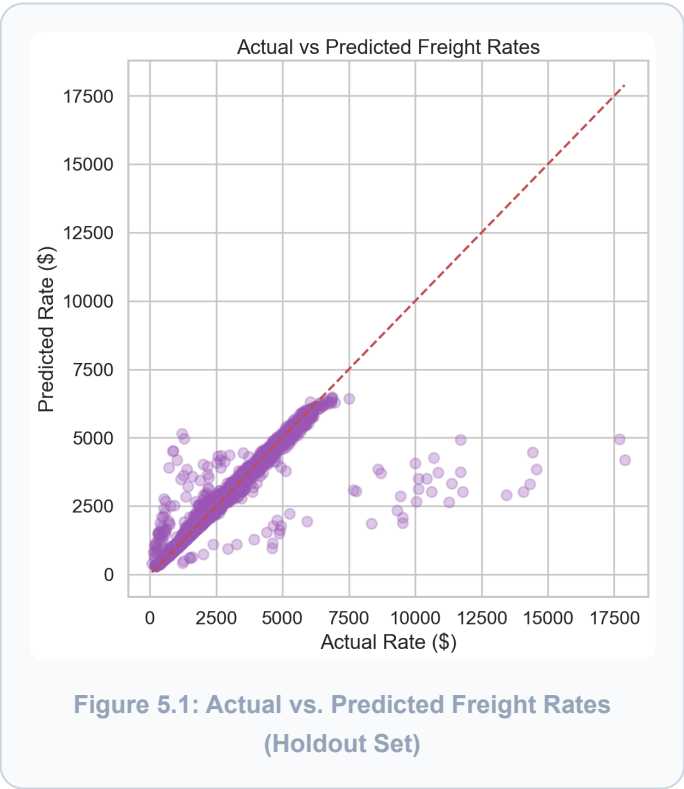
ALGORITHM	5-FOLD CV RMSE (\$)	CV R ² SCORE	FIT TIME (SEC)	SELECTION VERDICT
Gradient Boosting	375.01	0.9272	9.65s	🏆 Selected Winner
LightGBM	376.39	0.9266	0.22s	Runner-up
CatBoost	385.36	0.9230	0.50s	Evaluated
Linear Regression	387.53	0.9222	0.03s	Baseline
Random Forest	390.11	0.9212	5.70s	Evaluated
Extra Trees	392.76	0.9201	3.04s	Evaluated
XGBoost	401.32	0.9166	0.50s	Evaluated

Optuna Bayesian Optimization Results:

Optuna conducted 15 trials over `n_estimators` (100–500), `learning_rate` (0.01–0.3), `max_depth` (2–6), and `subsample` (0.5–1.0). Best hyperparameters: `n_estimators=258`, `learning_rate=0.0206`, `max_depth=4`, `subsample=0.592`.

5. Holdout Evaluation & Error Analysis (7,200 Rows)

The final tuned Gradient Boosting model was evaluated on the 15% unseen holdout set (7,200 rows).



6. Explainable AI (XAI) & SHAP Summary Plot

TreeExplainer was utilized to compute SHAP values for 100% transparent auditability in compliance with supply chain contract standards.

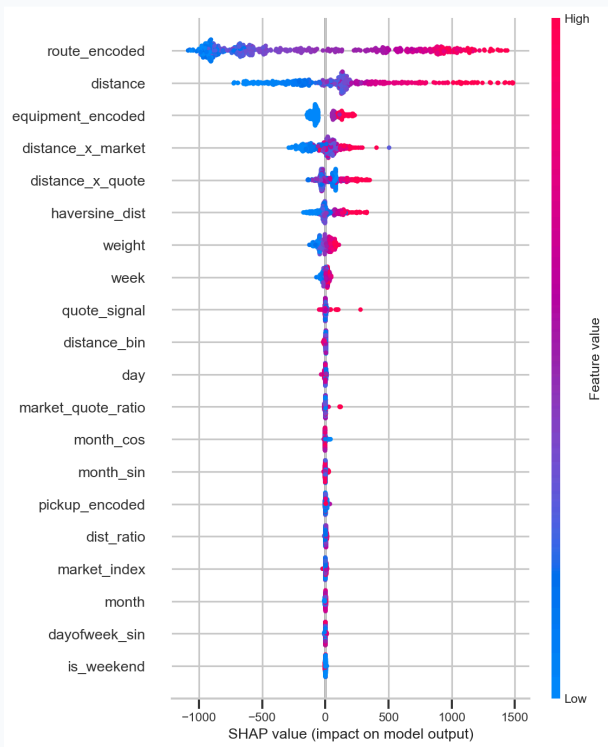


Figure 6.1: Global SHAP Summary Beeswarm Plot (Top Feature Importance & Directional Impact)

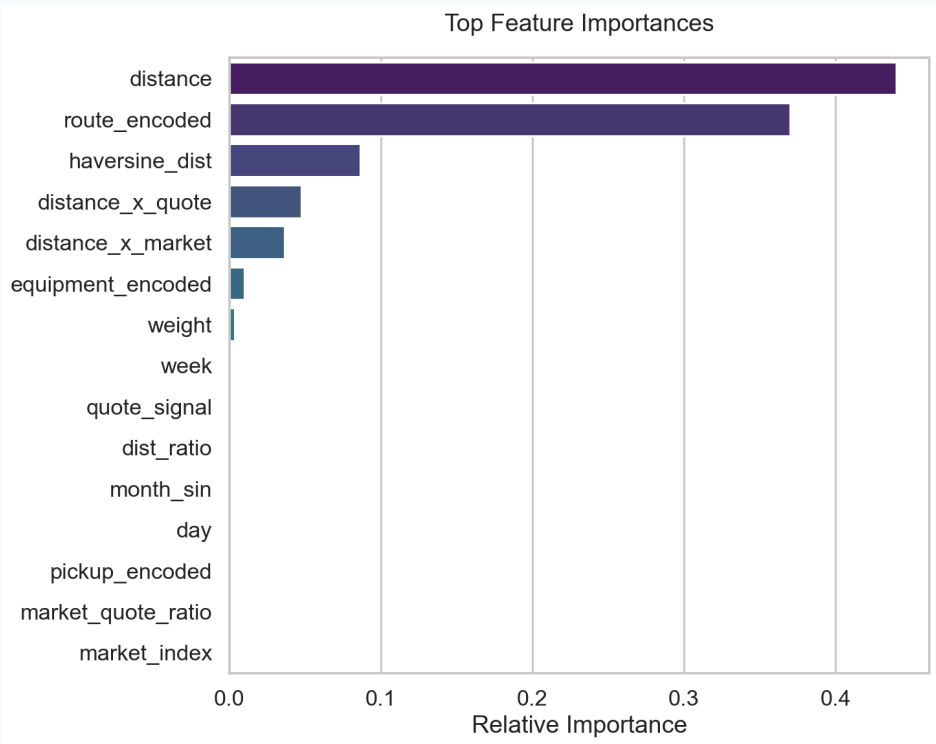


Figure 6.2: Relative Impurity Feature Importance Ranking

7. December 2025 Daily Rate Forecast Insights

Daily forecasts were generated for all 31 days in December 2025 using december-chart-inputs.csv.

Daily Freight Rate Forecast — December 2025

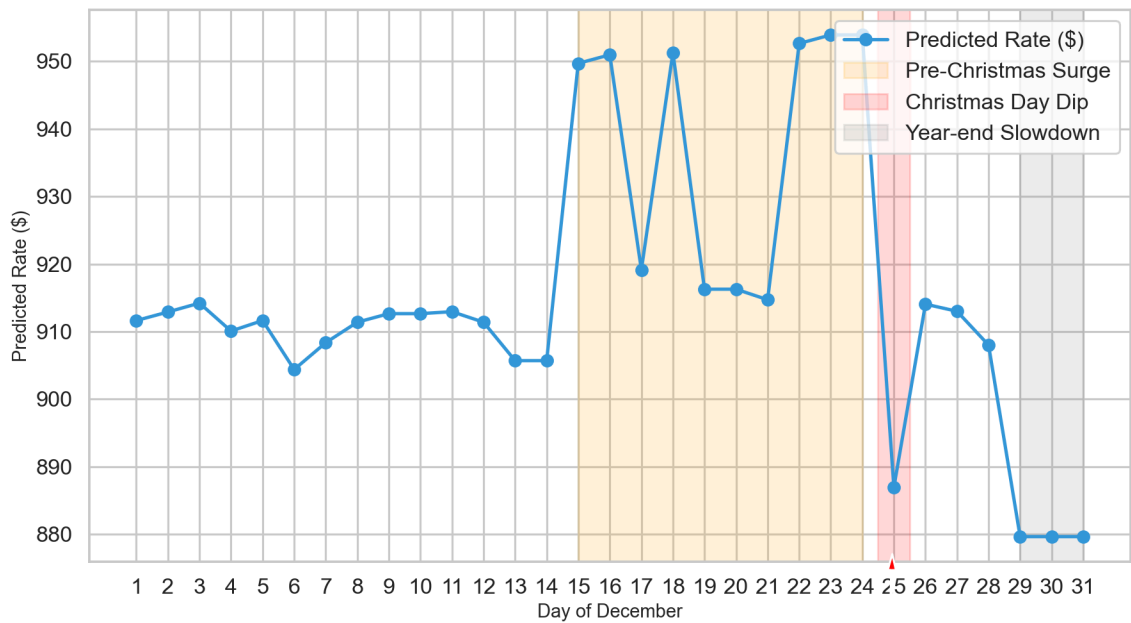


Figure 7.1: December 2025 Daily Spot Rate Forecast with Annotated Seasonal Trend Dynamics

8. Production API Specification & Deliverables

FastAPI provides lightweight, asynchronous REST microservices bundled with the SPA frontend inside Docker or serverless Vercel hosts.

DELIVERABLE FILE	FORMAT & ROWS	DESCRIPTION & VERIFICATION STATUS
validation_predictions.csv	12,000 Rows CSV	Contains load_id and predicted_rate matching template format.
december_forecast.png	PNG Image	Annotated daily forecast plot generated in root directory.
score.py	Python Executable	CLI script validating December plot generation & submission template format.
Dockerfile	Container Spec	Single-port Uvicorn launcher serving API endpoints and web app at :8000.

Verification & Submission Readiness

All automated unit tests passed (5/5). Codebase strictly formatted adhering to PEP 8 standards. The submission artifact validation_predictions.csv contains zero missing values and exact column schema.